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INSTITUTE OF PROFESSIONAL CERTIFICATIONS

Introduction to Generative AI

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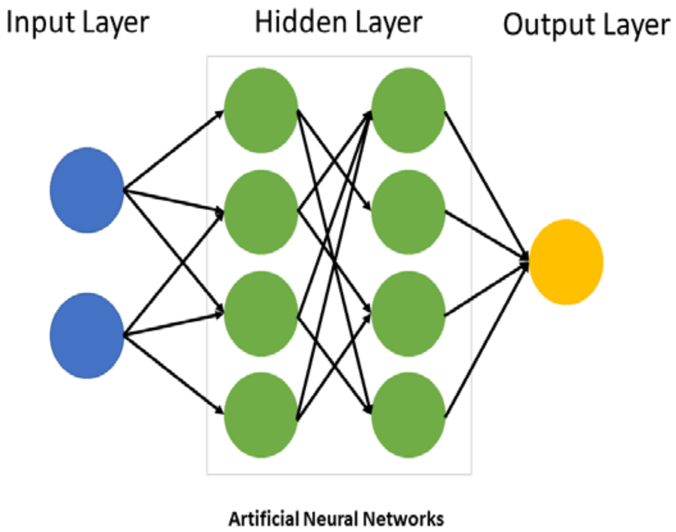
Why generative AI?

Amplifying Human Creativity: Not replacing, but enhancing human potential.

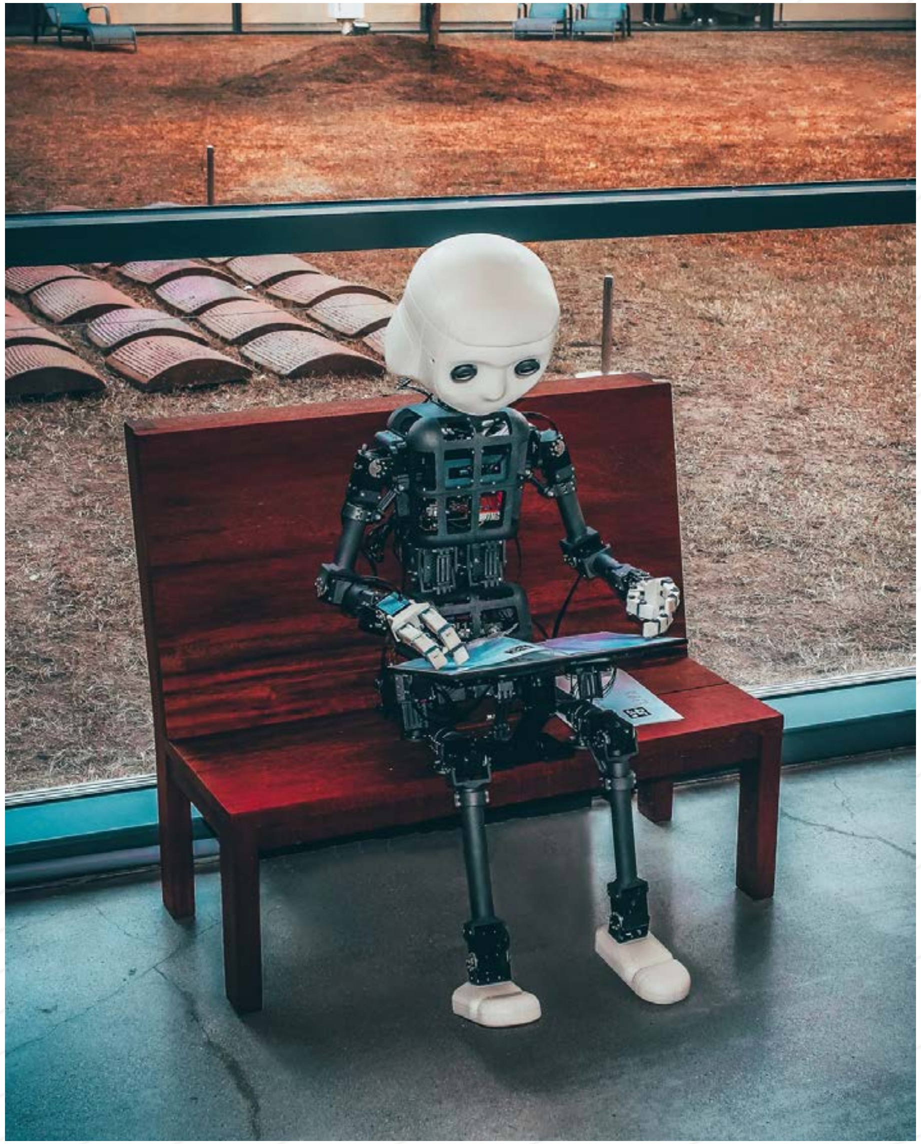
Efficiency and Innovation: Generate countless variations rapidly.

Beyond the Arts: Solutions for diverse industries from healthcare to finance



Artificial intelligence

Artificial Intelligence (AI) is the simulation of human intelligence processes by machines, especially computer systems.



Timeline of AI to Generative AI

Basic AI - 1950

Coined by John McCarthy in 1955
1956 Darmouth conference

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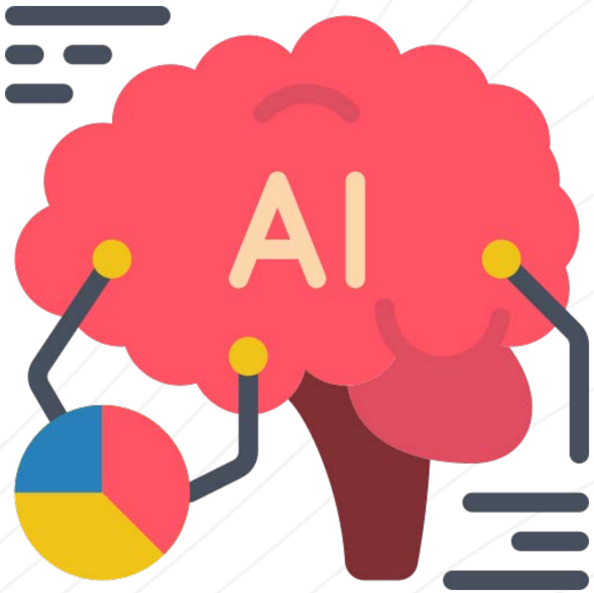
Generative AI - Late 2010 - Present

2014 - Generative Adversarial networks (GANa)

AI and its basic mission

AI is the capability of machines to imitate intelligent human behavior.

- Simulating Human Intelligence
- Problem Solving & Decision Making
- Learning & Adaptation
- Interaction with the Environment

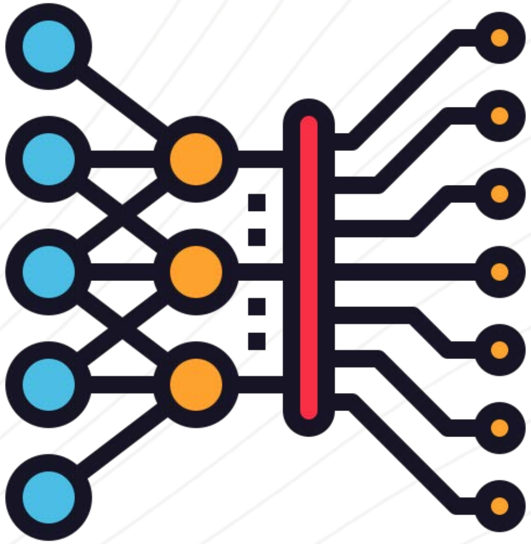


Machine Learning - Teaching Machines

ML allows computers to learn from data without being explicitly programmed.

“Classic or “non-deep” machine learning depends on human intervention to allow a computer system to identify patterns, learn, perform specific tasks and provide accurate results. Human experts determine the hierarchy of features to understand the differences between data inputs, usually requiring more structured data to learn.”

https://www.ibm.com/blog/ai-vs-machine-learning-vs-deep-learning-vs-neural-networks/



Deep Learning - The Deep Dive

Deep Learning uses layered neural networks

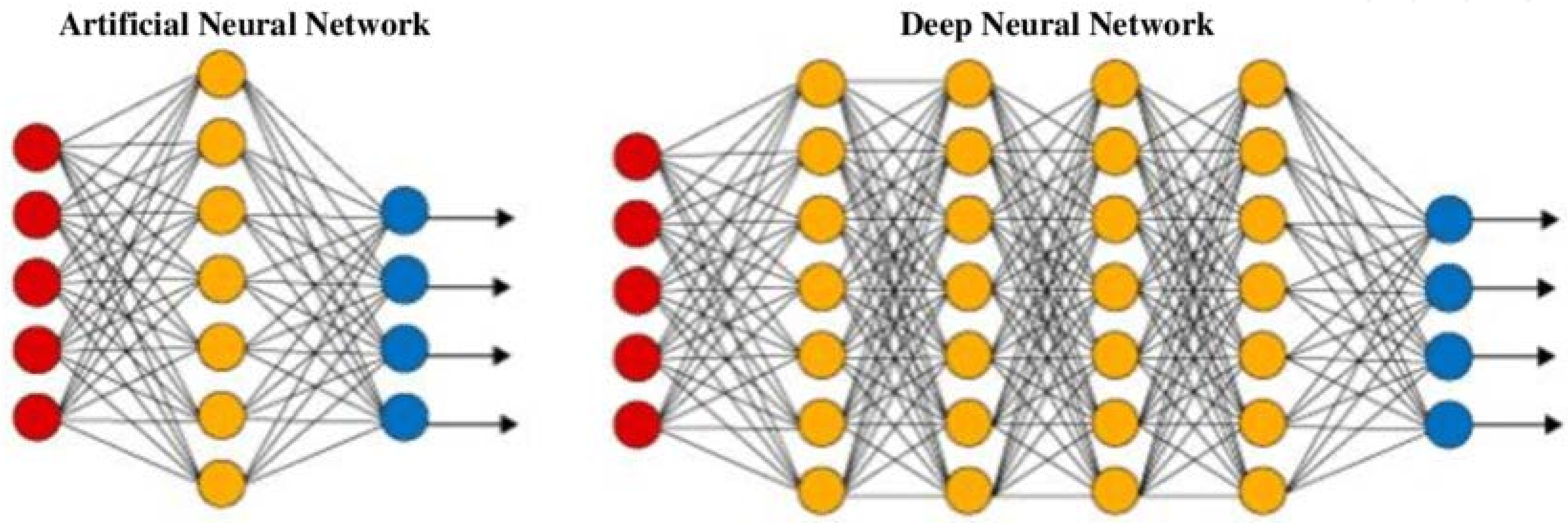


Image courtesy
deep-neural-net